INCH-POUND

MIL-DTL-5003F
5 June 2020
SUPERSEDING
MIL-DTL-5003E
21 June 2010

DETAIL SPECIFICATION

FLOODLIGHT SET, ELECTRIC, TRIPOD-MOUNTED, FOUR 150-WATT FLOODLIGHTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a tripod-mounted electric floodlight set designed for use in areas where explosion proof type illumination equipment is not required (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to DLA Aviation - VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616, or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil/.

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FEDERAL SPECIFICATION

TT-P-1757	- Primer Coating, ALKYD Base, One Component
W-C-596	- Connector, Electrical, Power, General Specification for

COMMERCIAL ITEM DESCRIPTIONS

A-A-55822	- Chests, Plywood
A-A-60006	- Light, Extension, Electrical

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-3432	- Cables, (Power and Special Purpose) and Wire, Electrical (300 and 600 Volts)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-171	- Finishing of Metal and Wood Surfaces
MIL-STD-130	- Identification Marking of U.S. Military Property
MIL-STD-889	- Dissimilar Metals
MIL-STD-1472	- Human Engineering
MIL-STD-1916	- DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other government documents. The following other government document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

Rule 442	- Usage of Solvents

(Copies of this document are available online at https://ww2.arb.ca.gov/.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ H1331	- Zero Acceptance Number Sampling Plans

(Copies of this document are available online at https://asq.org/.)

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SAE INTERNATIONAL

SAE AS12500	- Corrosion Prevention and Deterioration Control in Electronic Components and Assemblies
SAE AMS-QQ-P-416	- Plating, Cadmium (Electrodeposited)

(Copies of this document are available from https://www.sae.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Description. The tripod-mounted, electric floodlight set (herein after referred to as "lamp assembly") shall be as shown in figures 1 through 5 and as specified herein.

3.2.1 Figures. No deviations from the prescribed dimensions or tolerances are permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the figures to insure correct fit, assembly, and operation of the lamp assembly. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production items shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The lamp assembly shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the lamp assembly may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated

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representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the lamp assembly may be newly fabricated from recovered materials to the maximum extent practicable, provided the lamp assembly produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the lamp assembly.

3.4 Operation. The lamp assembly connected to a 120 volts alternating current (VAC), 50/60 Hertz (Hz) power source, the lamp cords and extension light plugged into the outlet box, and the switches in the "ON" position shall energize all the lamps. The lamp assembly shall show no signs of electrical or mechanical malfunction or damage as a result of the stand being erected to its maximum position and retracted to its carrying position.

3.5 Operational reliability. The lamp assembly shall withstand without malfunction, damage or permanent deformation 100 operational cycles when tested as specified in 4.4.2.3.

3.6 Electrical cable, plugs, and receptacles. The electrical cable shall conform to MIL-DTL-3432 as specified for low temperature, heat resistant cable (IS-L for insulation and JS-L for jacket). The connector plugs and connector receptacles shall conform to W-C-596. A protective boot shall be provided with each twist-lock connector plug which shall provide a raintight seal with the cable. All other connector plugs shall be molded to the cable.

3.7 Lamp-head assembly. The lamp-head assembly shall be as shown in figure 1 and shall be provided with a positive means to lock the assembly to the mast. The lamp-head housing cover shall be removable to provide access to the switch and wiring. All electrical wiring shall be insulated from the metal components of the housing. The electrical cable shall be a 3-conductor, 18 American wire gauge (AWG), light-duty cable, 12.5 +0.5/-0.0 feet long, of the type specified in 3.6, terminated at one end with a 3-prong, polarized, twist-lock, connector plug for connecting to the electrical connector box. The cable grounding conductor shall be grounded to the noncurrent carrying metal portion of the lamp-head assembly. The electrical switch for turning the lamps "ON" and "OFF" shall be a flush-mounted, weatherproof type.

3.8 Stand. The stand shall consist of a base, a mast, and a crossarm.

3.8.1 Base. The base shall be as shown in figures 2 and 3 and shall have legs that can be folded as shown in figure 4 for shipment or transport and locked in place during use. A strap shall be fastened to the crossarm for holding the legs and crossarms when in the retracted position.

3.8.1.1 Pegs and casters. Three-inch pegs and casters with step-on wheel brakes shall be provided to support the stand assembly. They shall be interchangeable.

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3.8.2 Mast. The mast shall be as shown in figure 2. The telescoping sections shall be provided with stops for preventing section from being detached from their mating tubes. T-type hand screws and shoes, or other comparable means, shall be provided for clamping the telescopic sections in position.

3.8.3 Crossarms. The crossarms shall be constructed to allow selective elevating and clamping of the arm along the lower section of the mast, and the arms shall be held in place by the same type of clamp that is used on the telescoping sections. Crossarms shall be fabricated to permit folding downward and parallel to the main tube for shipment and locking in place during use. One swivel lampholder shall be mounted on each crossarm. Each lampholder shall be equipped with 10 +0.5/-0.0 feet of 3-conductor, 18 AWG, light-duty electrical cable, of the type specified in 3.6, terminated at one end with a 3-prong, polarized, twist-lock connector plug for connecting to the electrical connector box. The cable grounding conductor shall be grounded to the noncurrent carrying metal portion of the lamp-head assembly.

3.9 Swivel lampholder. The swivel lampholder shall be weatherproof and equipped with a wire guard as shown in figure 5. The electrical cord length shall be as specified in 3.7 and 3.8.3, as applicable (see figures 1 and 2). The socket in the lampholder shall not fail when a 60 inch-pound torque is applied to the lamp installed in the socket. The following information shall be permanently and legibly marked on the lamp-head adjacent to the switch: "CAUTION-TURN OFF BEFORE REPLACING LAMPS. USE PAR-38 LAMP WITH MEDIUM SCREW SKIRTED BASE."

3.10 Lamps. Lamps shall be furnished by the supplier as part of the complete assembly, and an equal number of replacement lamps shall also be furnished. The portable carrying chest (see 3.13) shall contain eight 120 volt (V), 150 watt (W), medium screw skirted base, PAR-38 floodlights and two standard 120 V, 100 W, medium screw base lamps for extension light.

3.11 Electrical connector box. The electrical connector box shall be a weatherproof type containing six weatherproof outlets and three weatherproof switches. One switch shall control the two lamphead outlets. The second switch shall control the two crossarm light outlets and the third switch shall control the two outlets for the extension lights. The outlet shall be a 2-pole, 3-wire locking, grounding type in accordance with W-C-596 and shall match the plugs of the lamphead, crossarm, and extension lights respectively. The outlets and switches shall be labeled indicating which lights they control. The outlets shall be spaced sufficiently permitting all six outlets to be used simultaneously. The connector box shall be provided with 100 +0.5/-0.0 feet of 12 AWG, medium-duty, 3-conductor electrical cable, of the type specified in 3.6, terminated at one end with a polarized, heavy-duty, outdoor-service-type, molded connector plug having two parallel contacts and one ground connector contact. The cable grounding conductor shall be grounded to the mast.

3.12 Extension light. One portable extension light equipped with 50 feet of 18 AWG, medium-duty, 3-conductor electrical cable of the type specified in 3.6 shall be furnished. The portable extension light shall be in accordance with A-A-60006, type III, class 2, except for the type of cable and connector plug. The cable shall be terminated at one end with a 2-pole,

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3-wire, twist-lock grounding connector plug in accordance with W-C-596 for connecting to the electrical connector box.

3.13 Carrying chest. A portable carrying chest shall be provided for storing and transporting the lamp assembly and extension light. It shall be constructed in accordance with A-A-55822 of 3/8 inch thick plywood. Dimensional details fixing the location of the fixtures shall be determined by the contractor. Each lamp assembly, complete with replacement lamps (see 3.10), shall be packed in the carrying chest. Each component, part, or subassembly shall be secured within the carrying chest to prevent movement and breakage during handling and shipping. An 8 by 10 inch loading plan shall be provided.

3.14 Human factors engineering. The lamp assembly shall comply with the criteria of MIL-STD-1472. Special emphasis shall be given, but shall not be limited to, general requirements, controls, labeling, design for maintainer, and hazards and safety.

3.15 Fungus and moisture resistance. When specified (see 6.2), the electrical circuitry, including all components and connections, except as specified below, shall be treated for the effects of fungus growth and moisture as specified in SAE-AS12500.

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces. Surfaces shall be previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.15.1 Composition. All applications of varnish shall fully comply with SCAQMD Rule 442.

3.16 Finishes.

3.16.1 Mast, legs and crossarms. All exterior surfaces of the welded tubing made of other than corrosion-resistant metal shall be cadmium plated in accordance with SAE AMS-QQ-P-416, class 1, type II, or equivalent corrosion-resistant coating with equivalent thickness. Interior surfaces of the tubing made of other than corrosion resistant metal shall be coated with a corrosion-inhibiting primer in accordance with TT-P-1757, Type 1, Class N, and Color T unless otherwise specified for a different color.

3.16.2 Unpainted surfaces. The sliding portions of the lamp assembly and the glass section of the floodlight housing shall be cleaned and treated, but shall not be painted.

3.16.3 Treatment and painting. The lamp assembly shall be treated and painted in

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accordance with MIL-STD-171, color as specified (see 6.2).

3.17 Identification marking. Assemblies and parts shall be identified in accordance with MIL-STD-130, method optional.

3.18 Workmanship. The lamp assembly shall be free from defects such as chips, burrs, loose electrical connections, bent tubing and misaligned components.

3.19 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. Unless otherwise specified (see 6.2), two first article lamp assemblies shall be examined as specified in 4.4.1 and subjected to the tests specified in table I. Presence of one or more defects or failure of any test shall be cause for rejection.

TABLE I. Test schedule.

Test	First article	Conformance		Test paragraph	Requirement paragraph
		Individual	Sample		
Operation	X	X	-	4.4.2.2	3.4
Examination	X	X	-	4.4.1	See 4.4.1
Operation reliability	X	-	X	4.4.2.3	3.5, 3.9
Rainfall	X	-	-	4.4.2.4	3.7, 3.9, 3.11

4.3 Conformance inspection.

4.3.1 Sampling. Unless otherwise specified in the contract (see 6.2), sampling for examination and tests shall be in accordance with MIL-STD-1916, verification level (VL) IV or ASQ H1331, Index Value of 1.0.

4.3.2 Examination. Unless otherwise specified in the contract (see 6.2), sample selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.3.3 Tests.

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4.3.3.1 Individual. Each lamp assembly shall be subjected to the test specified in table I. Failure of the test shall be cause for rejection.

4.3.3.2 Samples. Samples selected in accordance with 4.4.1 shall be subjected to the test specified in table I. Failure of the test shall be cause for rejection.

4.4 Inspection procedure.

4.4.1 Examination. The lamp assembly shall be examined as specified herein for the following defects:

a. Lamp assembly not as specified in figures 1 through 5 (see 3.2).
b. Materials not as specified (see 3.3).
c. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable operating and storage environment (see 3.3.1).
d. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
e. Contractor does not have documentation available for identification of materials, material finishes, or treatments (see 3.3.1.2).
f. Used, rebuilt or remanufactured components, pieces, and parts incorporated in the lamp assembly (see 3.3.2).
g. Components not as specified (see 3.6, 3.7, 3.8, 3.9, 3.10, 3.11, 3.12, and 3.13).
h. Human Factors Engineering not as specified (see 3.14).
i. Fungus and moisture resistance treatment not as specified (see 3.15).
j. Finishes not as specified (see 3.16).
k. Identification marking not as specified (see 3.17).
l. Workmanship not as specified (see 3.18).

4.4.2 Tests.

4.4.2.1 Test conditions. Unless otherwise specified herein, tests may be performed without shelter and at the climatic conditions existing at the place of test. The lamp assembly shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test.

4.4.2.2 Operation. Connect the lamp assembly to a 120 VAC, 50/60 Hz power source. Plug the lamp cords and extension light into the outlet box and place the switches in the "ON" position. Erect the mast assembly from its retracted position to its maximum height and lock each mast section rigid and straight. Actuate the switches on the control box from the "OFF" to the "ON" position five times for switch control of lamp operation. Non-operation of the lamps (with exception of normal filament burn-out) or malfunction of either electrical switch or parts of the mast shall constitute failure of this test.

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4.4.2.3 Operational reliability. Test the lamp assembly in accordance with 4.4.2.2 except that the mast shall be extended and collapsed 100 times and the electrical switches shall be actuated from the "OFF" to the "ON" position 100 times. Apply 60 inch-pounds of torque to a lamp installed in one of the sockets. Nonconformance to 3.5 or 3.9 shall constitute failure of this test.

4.4.2.4 Rainfall. With the mast erected to normal operation position and the projector lamps operating, subject the lamp assembly to a simulated rainfall of 4 ±1 inches per hour for 1 hour. After exposure to the rain, examine the lamp assembly. Evidence of water penetration into the electrical components or malfunctioning of the lamps shall constitute failure of this test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The lamp assembly covered by this specification is intended to illuminate aircraft for maintenance operations in exposed locations under all weather conditions.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article is not required for inspection and approval (see 3.1, 4.2, and 6.3).
c. When fungus and moisture resistance treatment is required (see 3.15).
d. Color of paint to be specified (see 3.16.3).
e. VL/Index Value (see 4.3.1).
f. Packaging requirements (see 5.1).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more lamp assemblies. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products,

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who wish to rely on such production or test must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

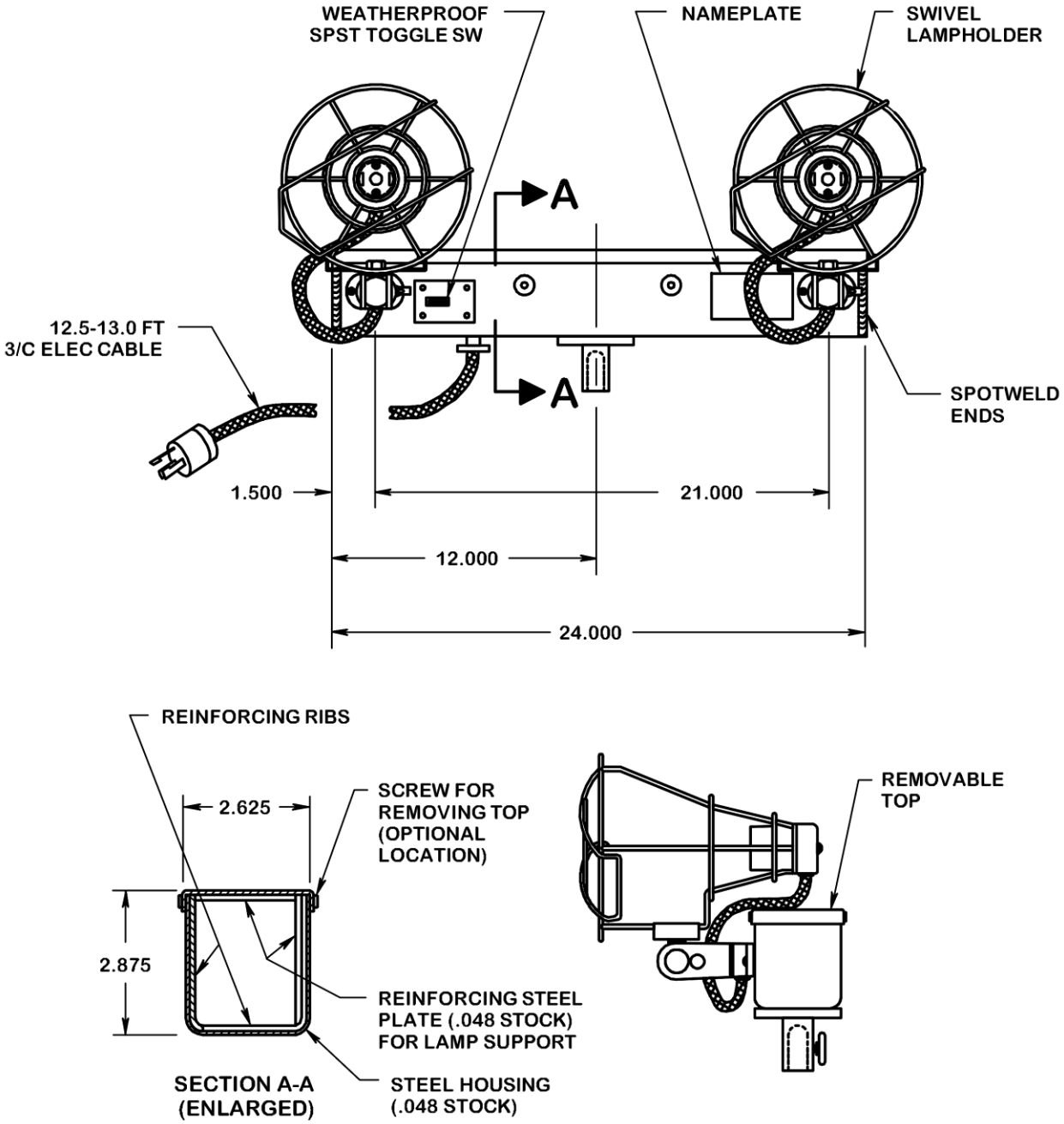
6.4 Safety Data Sheet (SDS). If the use of cadmium is not considered an article as specified in OSHA 3844, "Hazard Communication: Hazardous Classification Guidance for Manufacturers, Importers, and Employers", a SDS must be prepared and submitted in accordance with FED-STD-313 (obtain from https://assist.dla.mil). The SDS must also meet the requirements of 29 CFR 1910.1200. The 29 CFR 1910.1200 requires that the SDS for each hazardous chemical used in an operation must be readily available to personnel using the material.

6.5 Subject term (key word) listing.

Cadmium
Flood lamps
Lamp assembly
Lighting
Light stand
Portable

6.6 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

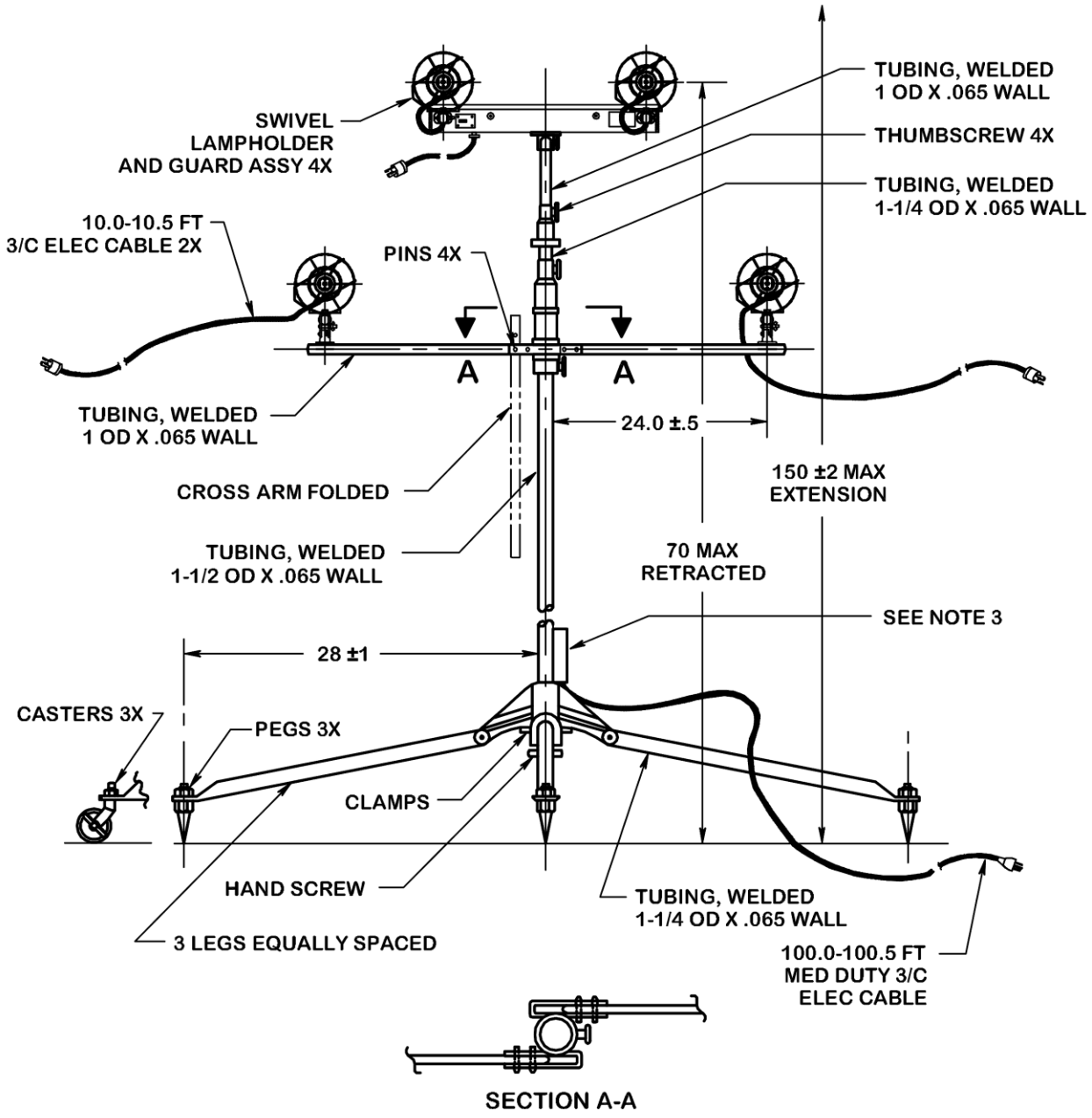
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NOTES:
1. Dimensions are in inches unless otherwise specified.
2. Unless otherwise shown, all tolerances are ±.062.

FIGURE 1. Lamp-head assembly.

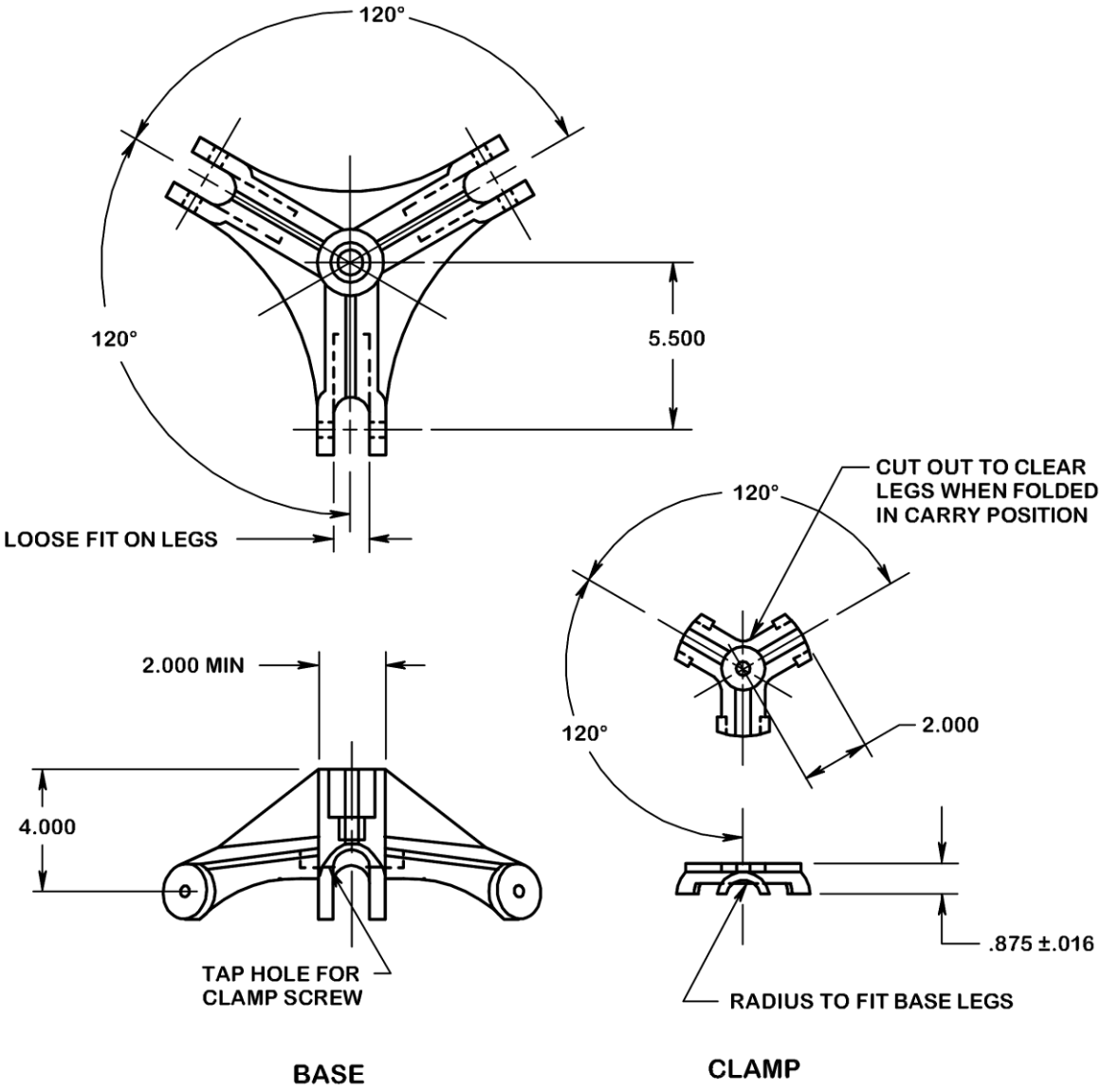
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NOTES:
1. Dimensions are in inches unless otherwise specified.
2. Tolerances: as noted.
3. Connector box with six electrical weatherproof connectors and two weatherproof switches.

FIGURE 2. Floodlight lamp assembly.

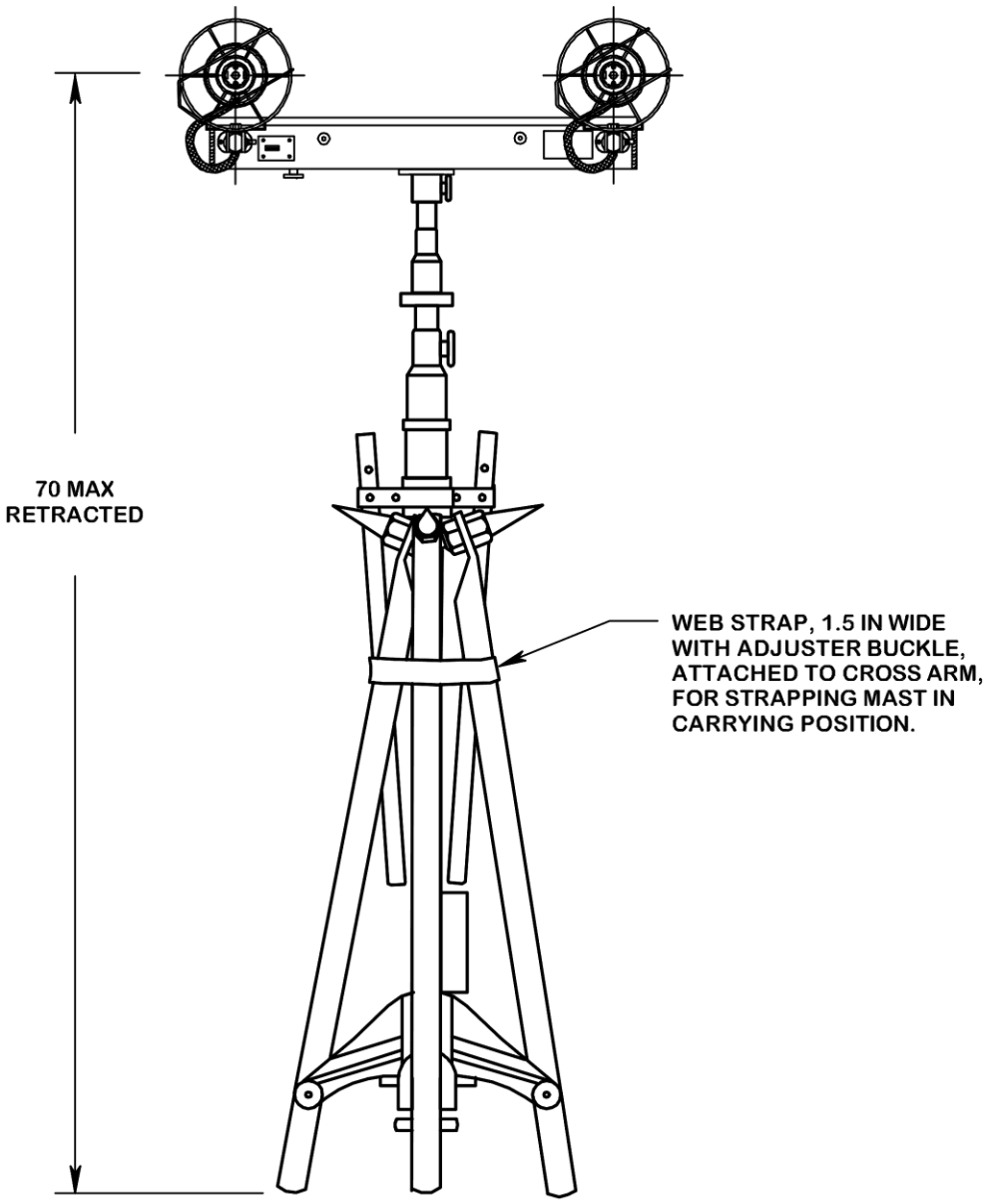
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NOTES:
1. Dimensions are in inches unless otherwise specified.
2. Tolerances: Dimensions ±.125. Angles ±1°.
3. Material: Malleable iron casting.

FIGURE 3. Base casting and leg clamp.

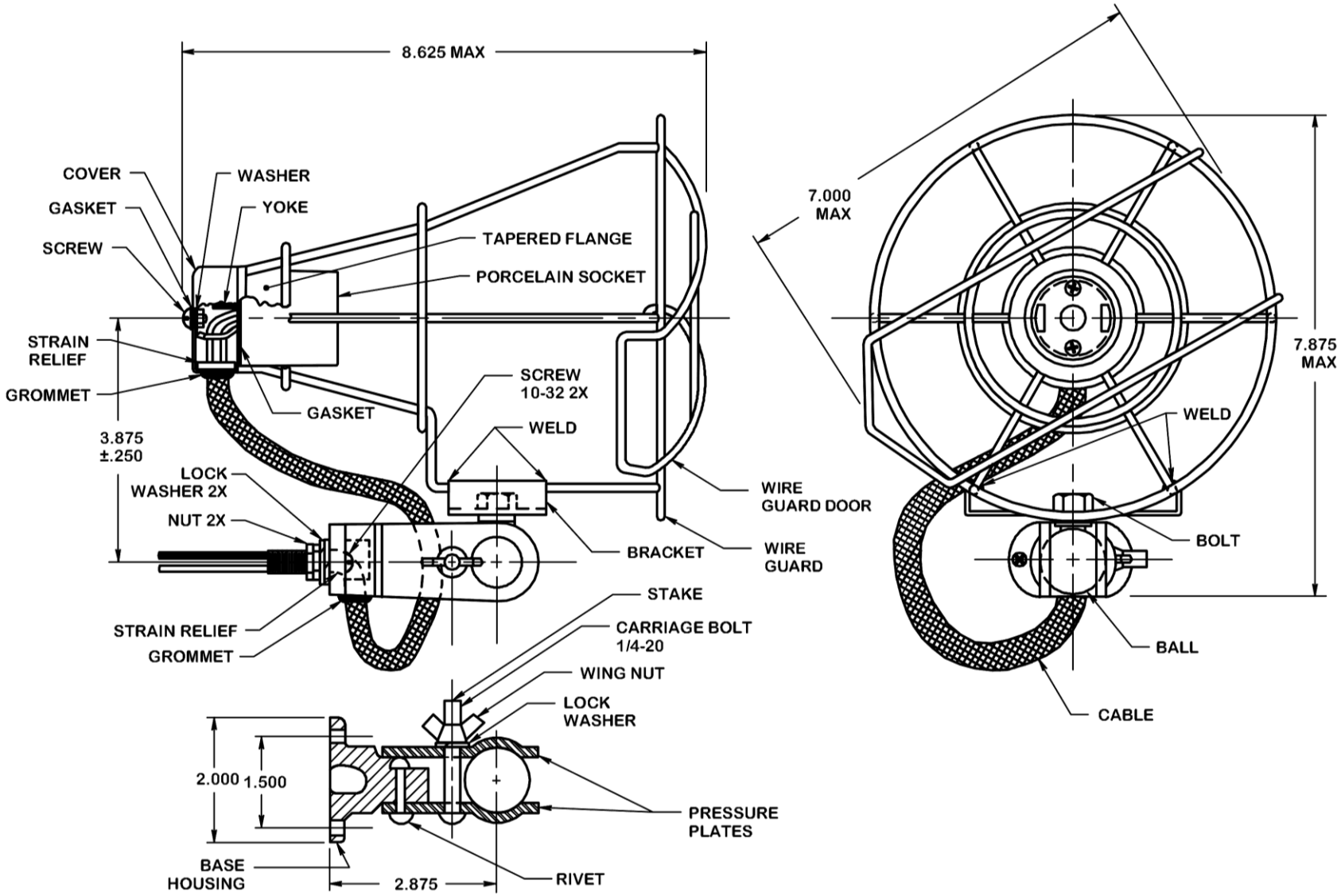
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NOTES:
1. Dimensions are in inches unless otherwise specified.

FIGURE 4. Mast in carrying position.

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NOTES:
1. Dimensions are in inches.
2. Unless otherwise shown, all tolerances are ±.016.

FIGURE 5. Swivel lampholder and guard assembly.

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